PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of Low-salt Therapy on Chronic Kidney Disease Outcomes: A meta-analysis
AUTHORS	Shi, Honghong; Su, Xiaole; Li, Chunfang; Guo, Wenjuan; Wang, Lihua

VERSION 1 – REVIEW

REVIEWER	Anothaisintawee, Thunyarat Ramathibodi Hospital, Mahidol University, Department of family medicine
REVIEW RETURNED	01-Apr-2021

GENERAL COMMENTS	In the abstract, the inclusion criteria is not clear. For instances, clinical study is very broad and the outcome of interest is not mentioned. The process of study selection is not mentioned in the abstract and the author should update searching. Using the term "renal failure" for the outcome is overclaimed because this outcome composed of rising serum creatinine, decline of eGFR and incidence of ESRD that the severity are very much different and might be misleading the audience. I suggest that the author should use the term "composite outcomes" instead. The definition of salt reduction is not explained in the method. Because the interventions of salt reduction might be varied in the different studies. Therefore, the author should state the definition and detail of salt reduction in each trial and also the measurement of salt intake in the observational studies. Publication bias is mentioned in the result but the method to assess publication bias is not stated in the method. The author should state clearly about the method of judging heterogeneity between studies. Using only P-value from heterogeneity test is not enough to consider heterogeneity between studies because this test has low power to detect the difference. Therefore heterogeneity might be present even the test is not significant. The author should explain more about how to assess risk of bias using the Newcastle and Ottawa such as how many domains and how to grade the quality of study. According to Figure 1, the number of studies that had no data are very high (75). Could you please explain, why these studies met

	your inclusion criteria but did not have enough data for extraction. Did you try to contact the corresponding authors? Studies in the review included the wide range of CKD patients (from CKD stage3 to ESRD). I think that these are too much different to combine as one group of patient. I suggest the author to stratify the analysis according to stage of CKD or separate ESRD patients. In the method, the author mentioned that the comparisons were different from studies. For examples, some used usual care as the comparator, while some studies, comparators were low salt diet or different level of salt diet. I think that is it appropriate to combine as one group?

REVIEWER	You, Zhiying University of Colorado Denver School of Medicine, Medicine
REVIEW RETURNED	18-Apr-2021

GENERAL COMMENTS	A very nice organized & written manuscript of a meta-analysis.

REVIEWER	Garneata, Liliana Nephrology II
REVIEW RETURNED	14-Jul-2021

GENERAL COMMENTS	The manuscript entitled “Effects of Low-salt Therapy on Chronic Kidney Disease Outcomes: A meta-analysis” addresses an interesting topic, looking at the effect on reduced dietary salt on several outcomes. It would be of interest for the BMC Nephrology readers. Several major points should be addressed before considering it for publication. Major points: 1. The authors used as primary kidney outcome measure the ”Renal failure event or kidney event”. However, this endpoint was not clearly defined. It should be defined even in the abstract. 2. How was the salt intake assessed in several studies ? 3. Are there data on diuretic use: type, dose. Are there any associations with the eGFR ? Minor points: 1. Low salt diet is dietary intervention, not a treatment itself. 2. The authors would like to temper down in some points, including conclusions: “Our meta-analysis demonstrates…”

VERSION 1 – AUTHOR RESPONSE

Reviewer #1:

1. In the abstract, the inclusion criteria is not clear. For instances, clinical study is very broad and the outcome of interest is not mentioned.
-Thank you for the comments. We have detailed the inclusion criteria in the abstract part (page 2, line 32-40).

“Eligibility criteria for selecting studies We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than 25% or 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD), decline rate in eGFR, changes in proteinuria, all-cause mortality events, cardiovascular (CV) events, change in SBP and DBP.”

2. The process of study selection is not mentioned in the abstract and the author should update searching.
-Thank you for your advice. We have updated the literature search. We identified 946 new potentially relevant records after updating the literature search. Finally, there are three relevant newly-published studies were included. Finally the total number of included studies is updated from 49 to 51. And the updated data have no significant change for the conclusion of outcomes in our study.

Revisions have been made in the Abstract (page 2, line 55-59, and page 3, line 4-15), Methods (page 5, line 36-41 of main document) and Results section (page 8, line 54-60, page 9, line4-18, line 51-60, page 10, line 4-58, page 11, line 4-17, line 50-59, page 12, line 4-17, and page 13, line 4-12), which now reads:

“We systematically searched MEDLINE by Ovid (from 1946 to July 2021), EMBASE (from 1966 to July 2021), and the Cochrane Library database (no date restriction) without language restriction. ”

“We identified 9948 potentially relevant records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. As shown in figure 1, Supplementary file 3 and Supplementary file 4, a total of 33 eligible studies including 13 RCTs and 19 cohort studies reported in 35 publications with 101077 participants, were included in this review. The median follow-up duration was 6 months (interquartile range 3 to 21). Participants were enrolled at an average age of 56 years, and male participants accounted for 60% of the total participants. The average eGFR of the included participants was 47.45 mL/min/1.73 m^2. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake.

“One RCT and 9 observational studies including 9578 patients with 2329 events compared the low-salt diet with the control in preventing renal composite outcomes events. As shown in figure 2, compared with the control, low-salt diet intervention produced a 34% reduction in the risk of renal composite outcomes events (RR: 0.72; 95% CI: 0.58, 0.89), with substantial heterogeneity (I^2 =87.2%, P <0.001). Subgroup analyses and meta-regression were conducted to discuss the source of

heterogeneity. No significant difference between RCTs and observational studies was found, while the effect sizes were greater in studies that enrolled patients aged < 58 years (P = 0.03) (table 1). On univariate meta-regression, no clear relationship between 24-hour urine sodium (P = 0.80) or baseline eGFR (P = 0.51) and the reduction in renal composite outcomes events was observed (Supplementary file 7, Supplementary file 8)."

"Five RCTs and 7 observational studies with a total of 3287 participants reported data regarding the effects of a low-salt diet on the rate of change in eGFR. Low-salt diet therapy did not produce an apparent beneficial effect on the rate of change in eGFR (MD:1.16; 95% CI: -2.02 to 4.33). There was evidence of significant heterogeneity across the included studies (I^2= 98.0%, P < 0.001) (figure 3). Subgroup analyses showed that there was no statistical heterogeneity according to prespecified characteristics (Supplementary file 9)."

"Data for the effects of low-salt diet intervention on all-cause mortality and CV events were available for 9 studies (two RCTs and seven observational studies) and 5 studies (one RCT and four observational studies), respectively. There were no clear effects of a low-salt diet on the risk of all-cause mortality (RR: 0.92; 95% CI: 0.53-1.46; 95997 participants and 4455 events) or CV events (RR: 1.01; 95% CI: 0.46-2.22; 93573 participants and 1989 events) compared with those in the control groups, with significant heterogeneity across the included studies (I^2 = 98.0%, 97.6% for all-cause mortality and CV events respectively, and P values for heterogeneity both less than 0.001) (figure 5)."

"The most notable exception was that the effect of low-salt diet therapy on the rate of change in eGFR became significant when studies with extreme outliers were omitted."

And we have described the process of study selection in the abstract (page 2, line 55-59).

"We identified 9948 potentially records. After removal of duplicates, we screened the titles and abstracts, and 230 publications were reviewed in full text. Thirty-three studies with 101077 participants were included."

3. Using the term "renal failure" for the outcome is overclaimed because this outcome composed of rising serum creatinine, decline of eGFR and incidence of ESRD that the severity are very much different and might be misleading the audience. I suggest that the author should use the term "composite outcomes" instead.
-Thank you for your comments, we have taken your advice. The term "renal failure" has been revised to term "renal composite outcomes".

4. The definition of salt reduction is not explained in the method. Because the interventions of salt reduction might be varied in the different studies. Therefore, the author should state the definition and detail of salt reduction in each trial and also the measurement of salt intake in the observational studies.

-Thank you for your comments. We have explained the definition of salt reduction in the method (page 6, line 9-15). And the definition and detail of salt reduction in each trial and also the measurement of salt intake in the observational studies were stated in the Supplementary file 3, Table S1.

“Salt reduction was defined as that recommended by guidelines sodium <2.3 g/day (<100 mmol of sodium, <6g of salt), or as defined by the authors of each study.”

Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9*	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 80-100 mmol/d	Sodium intake 80-100 mmol/d	24-h urinary sodium	Usual care	1.39c	143/88	NA
Fujiwara S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPA KUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5*	144/95	Non- industry

5. Publication bias is mentioned in the result but the method to assess publication bias is not stated in the method.

-Thank you for your comments. We have stated the method to assess publication bias in method section (page 8, line 30-33 of main document).

“Publication bias was assessed using Egger’s test and Begg’s test, and *P*-value < 0.05 indicated obvious publication bias.”

6. The author should state clearly about the method of judging heterogeneity between studies. Using only P-value from heterogeneity test is not enough to consider heterogeneity between studies because this test has low power to detect the difference. Therefore heterogeneity might be present even the test is not significant.

-Thank you for your comments, and we totally agree with you. Using only *P*-value from heterogeneity test is not enough to consider heterogeneity between studies. In the method section, we stated the method of judging heterogeneity between studies (page 8, line 25-29).

“Heterogeneity between studies was evaluated by using I^2 statistics or τ^2 statistics. The I^2 value of 25%, 50%, 75% respectively represent low, moderate, and high heterogeneity. All I^2 statistics were stated in the results part of our manuscript. And τ^2 statistics were stated in the supplementary file 11 (results of sensitivity analyses). ”

7. The author should explain more about how to assess risk of bias using the Newcastle and Ottawa such as how many domains and how to grade the quality of study.
-Thank you for your comments. We have explained more about how to assess risk of bias using the Newcastle and Ottawa such as how many domains and how to grade the quality of study in method section according to reference 12 Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *Eur J Epidemiol* 2010;25(9):603-5. (page 7, line 7-20).

“The NOS contains eight items, categorized into three dimensions, including selection of cohorts, comparability of cohorts, and assessment of outcome [12]. A star system is used to allow a semi-quantitative assessment of study quality, such that the highest quality studies are awarded a maximum of one star for each item with the exception of the item related to comparability that allows the assignment of two stars. The NOS ranges between zero up to nine stars. ”

8. According to Figure 1, the number of studies that had no data are very high (75). Could you please explain, why these studies met your inclusion criteria but did not have enough data for extraction. Did you try to contact the corresponding authors?
-Thank you for your comments. To avoid the final included studies were incomplete, after screened the titles and abstract, we included all potential studies that low-salt diet was given to adults with CKD. However, when we tried to find full text for these studies, we found 53 studies were conference abstract or protocol. And the main interventions of 11 studies were other drugs (such as ARB, ACER, vitamin D), meanwhile combined with low salt diet, so they actually did not met our inclusion. Main interventions of 4 stusies were other drugs (such as ARB, ACER, vitamin D), meanwhile combined with low salt diet or routine salt diet, but there were not enough data about the outcomes of different salt intake, and we have tried to contacted the corresponding authors, but we did not get the reply. Two studies enrolled in CKD and non-CKD patients. We could not get the data for CKD patents, and we tried to have tried to contacted the corresponding authors, but we did not get the reply. Three studies described the results using the urine sodium excretion by linear regression analysis, so we contacted the corresponding authors, but we did not get the reply.

9. Studies in the review included the wide range of CKD patients (from CKD stage3 to ESRD). I think that these are too much different to combine as one group of patient. I suggest the author to stratify the analysis according to stage of CKD or separate ESRD patients.
-Thank you for your advice. Because of the stages of CKD patients in the studies we included varies a lot, some are CKD 1-5, some are CKD 1-3, some are CKD 3-4, some are CKD 2-4, some are CKD 4-5, some are CKD 3-4. So it is difficult to stratify our analysis our outcomes by stage of CKD. So, we performed subgroup analyses according to the baseline eGFR of CKD patients. The results were showed in Table 1 and Supplementary file 9. And we have described this as a part of limitation in the discussion part (page 17, line 4-15).

10. In the method, the author mentioned that the comparisons were different from studies. For

examples, some used usual care as the comparator, while some studies, comparators were low salt diet or different level of salt diet. I think that is it appropriate to combine as one group?
-Thank you for your comments, we totally agree with you. The comparison different from studies. Heterogeneity may be introduced combining as one group. Clinical heterogeneity is an inevitable issue in every meta-analysis. In our study, there are different comparators, including usual-care or usual-diet, and different levels of sodium intake. Twelve studies were usual-care or usual-diet controlled, and twenty-one trials compared different levels of sodium intake. We conducted subgroup analysis according to the different comparators, and no statistical heterogeneity was found. See in Table 1, and Supplementary file 9, Table S5.

Reviewer #2:

Thank you for your comments.

Reviewer #3:

Major points:

1. The authors used as primary kidney outcome measure the ”Renal failure event or kidney event”. However, this endpoint was not clearly defined. It should be defined even in the abstract.
-Thank you for your advice. Combined with the advice of reviewer 1 we have revised “renal failure event” to “renal composite outcomes event”, and defined it in the abstract (page 2, line 35-37 of main document).

“We included RCTs and cohort studies that evaluated the effects of low-salt diet on the renal composite outcomes (more than 25% or 50% decline in eGFR during follow-up, doubling of serum creatinine or ESRD).....”

2. How was the salt intake assessed in several studies ?
-Thank you for your comments. Sodium intake was assessed using 24-h urine sodium (20 studies), urinary sodium/creatinine (4 studies), dietary recall and food frequency questionnaires (6 studies), analysis of the dietary protocol and analysis of urine (1 study) or modified balance formulas (2 studies). The details were sated in the Supplementary file 3.

Table S1 Characteristics of included trials and patients.

Study (Year)	Inclusion criteria	Size of study	Men (%)	Mean age, years	Type of study	Follow-up, months	Outcome	Treatment group	Definition and detail of salt reduction	Measurement of salt intake	Control group	Mean eGFR	Mean blood pressure (mmHg)	Funding Source
Mühlhauser I[18] (1996)	Insulin-dependent diabetic patients with nephropathy and high normal or mildly hypertensive blood pressure	16	75	35	Randomised parallel-group controlled trial	1	UPE, SBP, DBP	Reduce sodium intake to about 90 mmol/d	Reduce sodium intake to about 90 mmol/day	Salt intake were estimated by analysis of the dietary protocol and analysis of urine	Sodium intake about 190 mmol/d	87.9*	132/101	Non- industry
Cianciaruso B[19] (1998)	Ccr 10–40 mL/min	57	72	50	Cohort study	42.95	Rate of change in eGFR, UPE	Urinary sodium < 100 mmol/d	Urinary sodium < 100 mmol/d	24-h urinary sodium	Urinary sodium > 200 mmol/d	38	NA	NA
Keven K[20] (2006)	Patients underwent kidney transplantation and had stable allograft function	32	78	41	Randomised parallel-group controlled trial	3	SBP, DBP	Sodium intake 50-100 mmol/d	Sodium intake 50-100 mmol/d	24-h urinary sodium	Usual care	1.39c	145/88	NA
Fujimura S[21] (2010)	UACR ≥ 30 mg/g Cr	36	56	73	Cohort study	3	UACR, SBP, DBP	Intensive low-salt diet education	NA	salt intake was examined, using Excel Eiyokun (version 4.5, KENPA KUSHA, Tokyo, Japan) for the assessment of dietary intake by food frequency questionnaire.	Usual care	65.6	140/67	Non- industry
Yu W[22] (2011)	Immunoglobulin A nephropathy	43	53	36	Cohort study	0.25	UPE, SBP, DBP	Sodium intake 100mmol/d	Sodium intake 100mmol/d	24-h urinary sodium	Usual diet	70.2	118/75	NA
Vegter S[23] (2012)	CKD patients without diabetes	500	76	52	Cohort study	51	Renal failure	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine < 200 mmol/g	Urinary sodium/creatinine	Urinary sodium/creatinine≥ 200 mmol/g	42.5*	144/95	Non- industry

3. Are there data on diuretic use: type, dose. Are there any associations with the eGFR ?
-Thank you for your comments. Ten studies have the data about diuretic use, but only about the percentage of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There are no data about type and dose of diuretics, or the associations with the eGFR. We have described this in discussion part (page 14, line 41-54).

“Given the use of diuretic may influence the urine sodium excretion, and eGFR, and it would influence the accuracy of quantified salt intake. We extracted data about use of diuretic. Ten studies have the data about diuretic use in the study we included, but only about the percentages of patients using diuretics in the intervention and control group. The percentage of patients using diuretics was no difference in most studies. There were no data about type and dose of diuretics, or the associations with the eGFR.”

Minor points:

1. Low salt diet is dietary intervention, not a treatment itself.
-Thank you for your advice. We have revised the low salt diet therapy to low salt diet intervention.

2. The authors would like to temper down in some points, including conclusions: “Our meta-analysis demonstrates…”
-Thank you for your advice. We have tempered our points (page 13, line 25-30, and page 17, line 30-33).

“Our meta-analysis showed that compared with control groups, low-salt diet intervention seems to reduce renal composite outcomes events and lowered BP (SBP and DBP) in CKD patients.”

“In conclusion, our meta-analyses suggested that a low-salt diet may reduce the risk of renal composite outcomes events in patients with CKD.”

VERSION 2 – REVIEW

REVIEWER	Anothaisintawee, Thunyarat Ramathibodi Hospital, Mahidol University, Department of family medicine
REVIEW RETURNED	29-Sep-2021

GENERAL COMMENTS	Abstract Why did author not include systematic review in title and objective of the study? Introduction “According to data from the NHANE II study, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans” I wonder this sentence (in lines between 49-55) is correct or not? Please check. Methods The definition of renal composite outcomes is too broad. The author should provide the rationale for combining more than 25% with 50% decline in eGFR from baseline for this outcome in the methods and should mention this limitation in the discussion. Using only Egger test for assessing publication bias may be less sensitive to detect the publication bias. I suggest using the Funnel plot to further explore the publication bias. Results Forest plots of each outcome should be provided. In addition, intervention and comparator for RCTs and threshold of Na intake used to categorize low and high Na intake for cohort studies should be presented for each pooling. The data that used for pooling for each outcome should be presented. Discussion The authors should be discussed about the limitation of combining RCT with cohort studies and discuss more about the subgroup analysis according to study design. I have a concern about pooling evidence from RCT with evidence from cohort studies because cohort studies are prone to selection bias that is usually not found in RCT. Therefore, pooling the results from RCT and cohort studies together will bias the results of meta-analysis.

REVIEWER	Garneata, Liliana Nephrology II
REVIEW RETURNED	30-Aug-2021

GENERAL COMMENTS	The authors performed adequate changes to the manuscript, significantly improving it. I would recommend the paper for publication, considering the relevance for the readers of BMJ.

VERSION 2 – AUTHOR RESPONSE

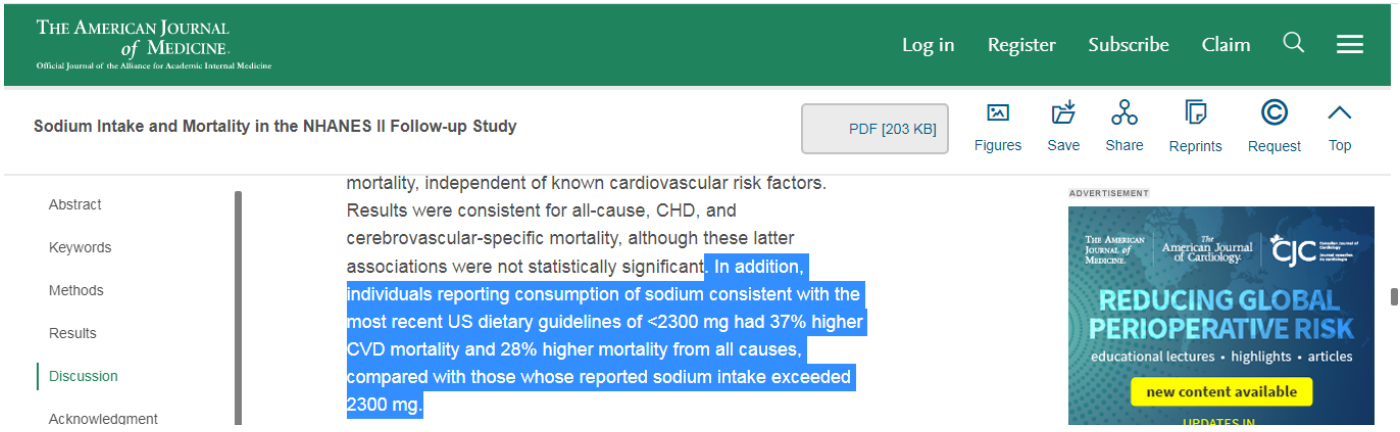
Reviewer #1:

1. Why did author not include systematic review in title and objective of the study?

-Thank you for your advice. We have included systematic review in the title and objective of the study.

2. “According to data from the NHANE II study, restriction of dietary sodium intake to <2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans” I wonder this sentence (in lines between 49-55) is correct or not? Please check.
-Thank you for your comments. We have checked the sentence, and found it is inappropriate. We have corrected it See in page 4 , line 49-57.

“According to data from the NHANE II study, compared with those whose sodium intake exceed 2.3g daily, restriction of dietary sodium intake to < 2.3 g daily resulted in increases of 37% in CV mortality and 28% in all-cause mortality over 13.7 years of follow-up among 7154 community-dwelling Americans.”



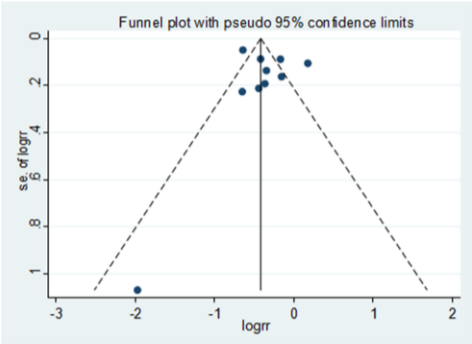
3. The definition of renal composite outcomes is too broad. The author should provide the rationale for combining more than 25% with 50% decline in eGFR from baseline for this outcome in the methods and should mention this limitation in the discussion.
-Thank you for your comments, we totally agree with you about the definition of renal composite outcome is too broad. To improve the study power, referring to the prior study, Coresh J, Turin TC, Matsushita K, et al. Decline in estimated glomerular filtration rate and subsequent risk of end-stage renal disease and mortality. JAMA. 2014;311(24):2518-2531, the surrogate outcomes for ESRD risk

of a reduction in eGFR >25% or 50% are defined as renal composite outcomes in the protocol of our study. After finishing the data extraction, we found that no study used the definition of >25 % reduction in eGFR. Now we have defined a reduction in eGFR > 50% as renal composite outcomes (Page 6, line 17-20).

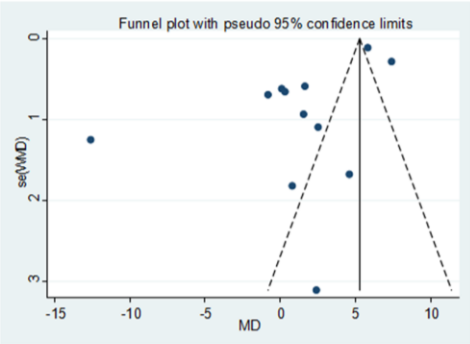
4. Using only Egger test for assessing publication bias may be less sensitive to detect the publication bias. I suggest using the Funnel plot to further explore the publication bias.

-Thank you for your advice, and we totally agree with you. We have used the Funnel plot to further explore the publication bias. The risk of publication bias was found to be significant for the rate of change in eGFR, proteinuria or albuminuria, all-cause mortality and cardiovascular events, and BP

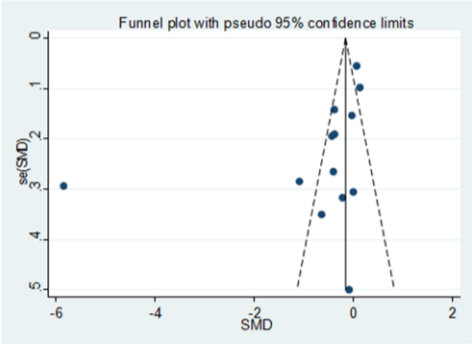
(Supplementary file 13) (Page 1



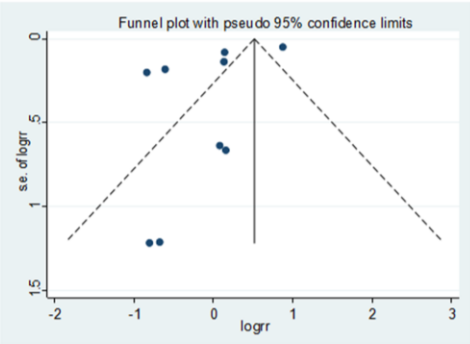
Funnel plot for publication bias on renal composite outcomes events.



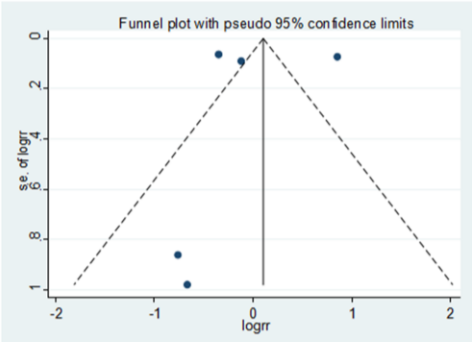
Funnel plot for publication bias on rate of change eGFR.



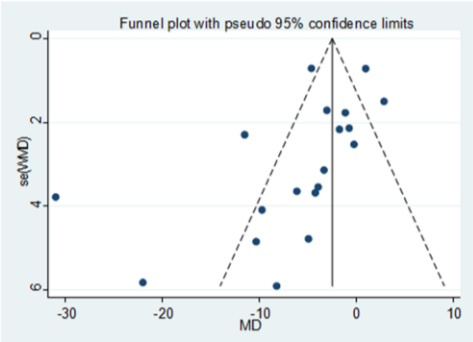
Funnel plot for publication bias on change in proteinuria or albuminuria.



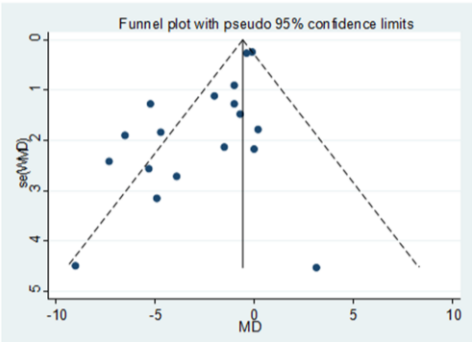
Funnel plot for publication bias on all-cause mortality events.



Funnel plot for publication bias on CV events.



Funnel plot for publication bias SBP.



Funnel plot for publication bias on DBP.

3, line 12-17).

Supplementary file 13 Funnel plot for publication bias on outcomes. CV, cardiovascular; DBP, diastolic blood pressure; MD, mean difference; RR, relative risk; SMD, standard mean difference; SBP, systolic blood pressure.

5. Forest plots of each outcome should be provided. In addition, intervention and comparator for RCTs and threshold of Na intake used to categorize low and high Na intake for cohort studies should be presented for each pooling.

The data that used for pooling for each outcome should be presented.

-Thank you for your comments. We have provided Forest plots of each outcome, and intervention and comparator for RCTs and threshold of Na intake used to categorize low and high Na intake for cohort studies. The data that used for pooling for each outcome were also presented in the Forest plots. (See in Figure 2, Figure 3, Figure 4, Figure 5, Supplementary file 10, Supplementary file 11).

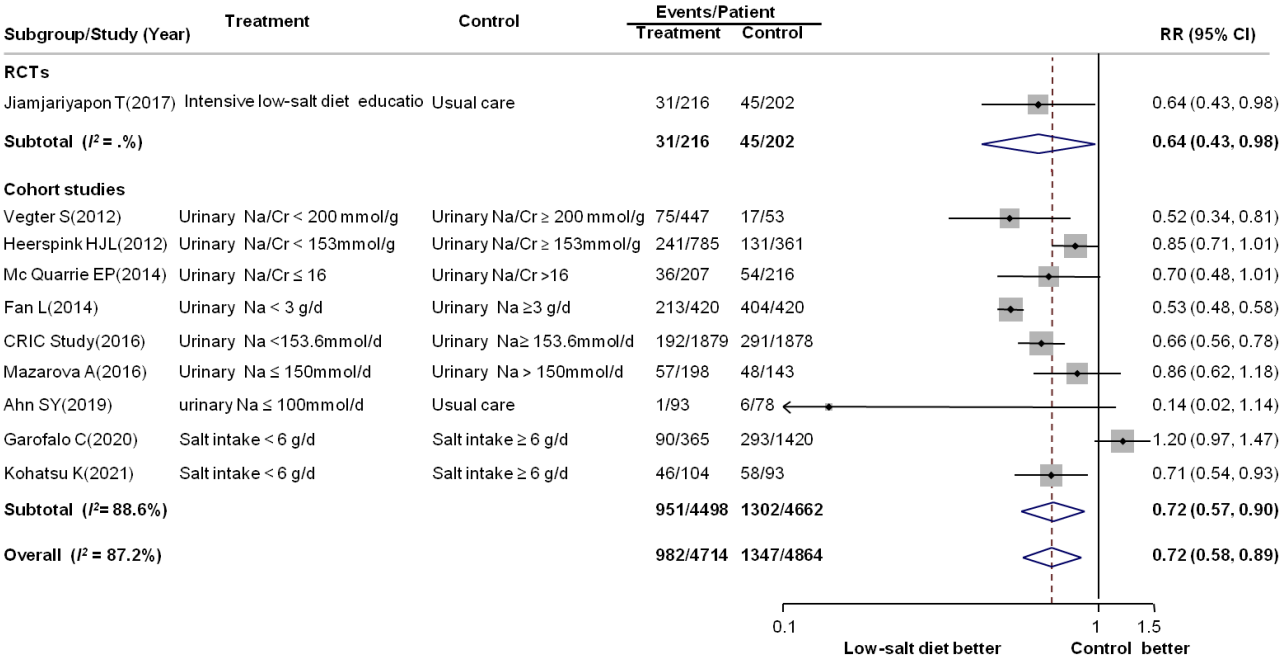


Figure 2 Forest plot for renal composite outcomes events. Renal composite outcomes events was defined as a more than 50% decline in eGFR from baseline during follow-up, doubling of serum creatinine or ESRD. CI, confidence interval; Cr, creatinine; Na, sodium; RR, relative risk.

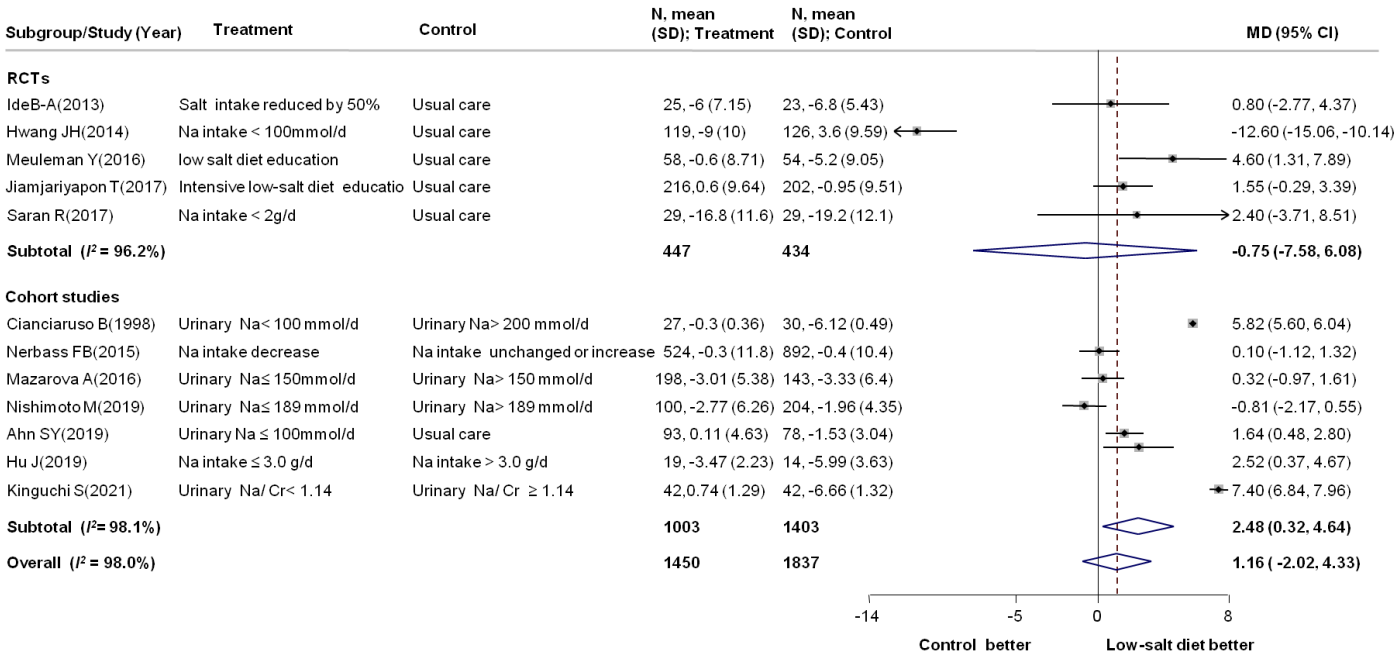


Figure 3 Forest plot for rate of change in estimated glomerular filtration rate (eGFR). CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SD, standard deviation.

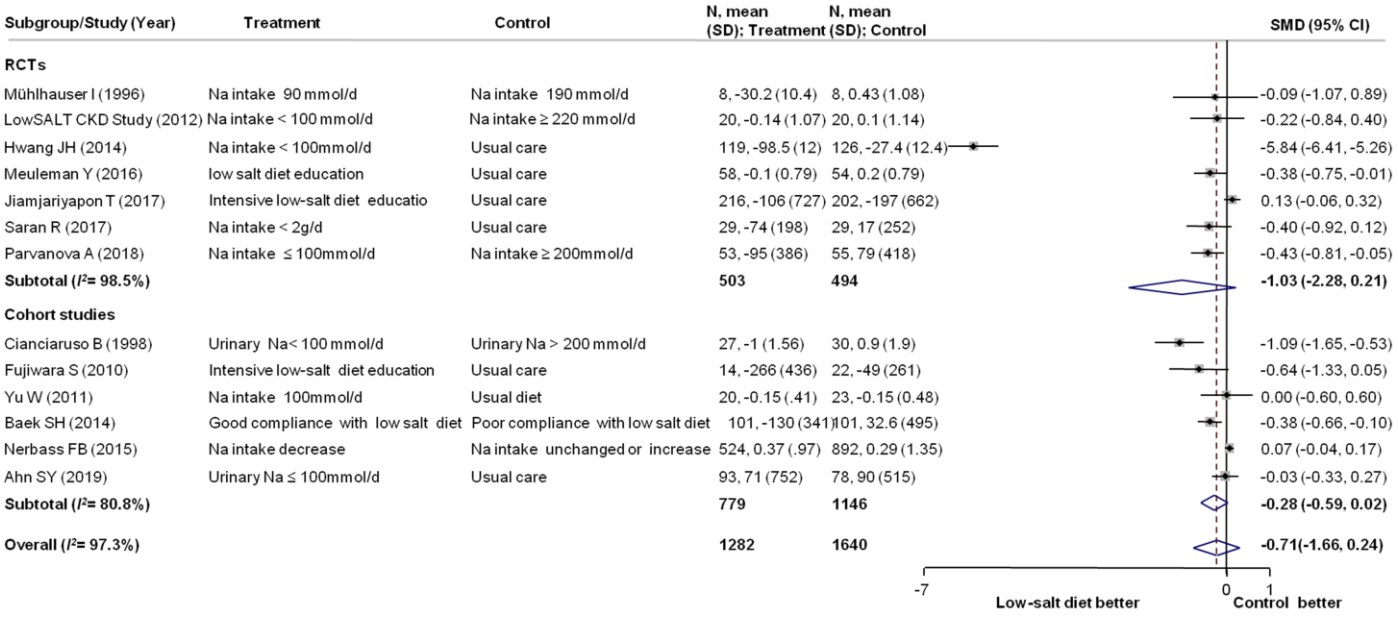


Figure 4 Forest plot for the change in proteinuria or albuminuria. CI, confidence interval; Cr, creatinine; Na, sodium; SD, standard deviation; SMD, standard mean difference.

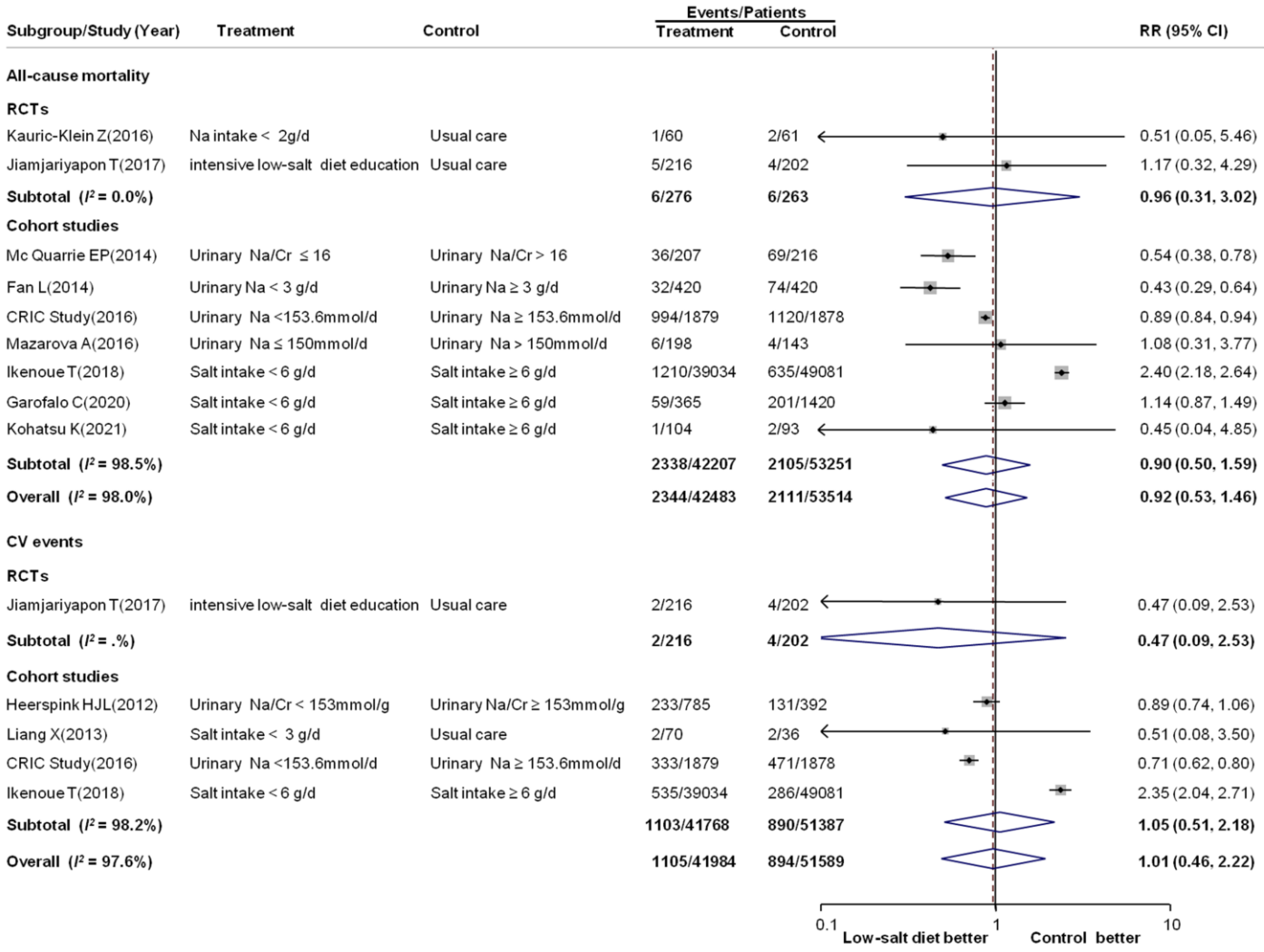
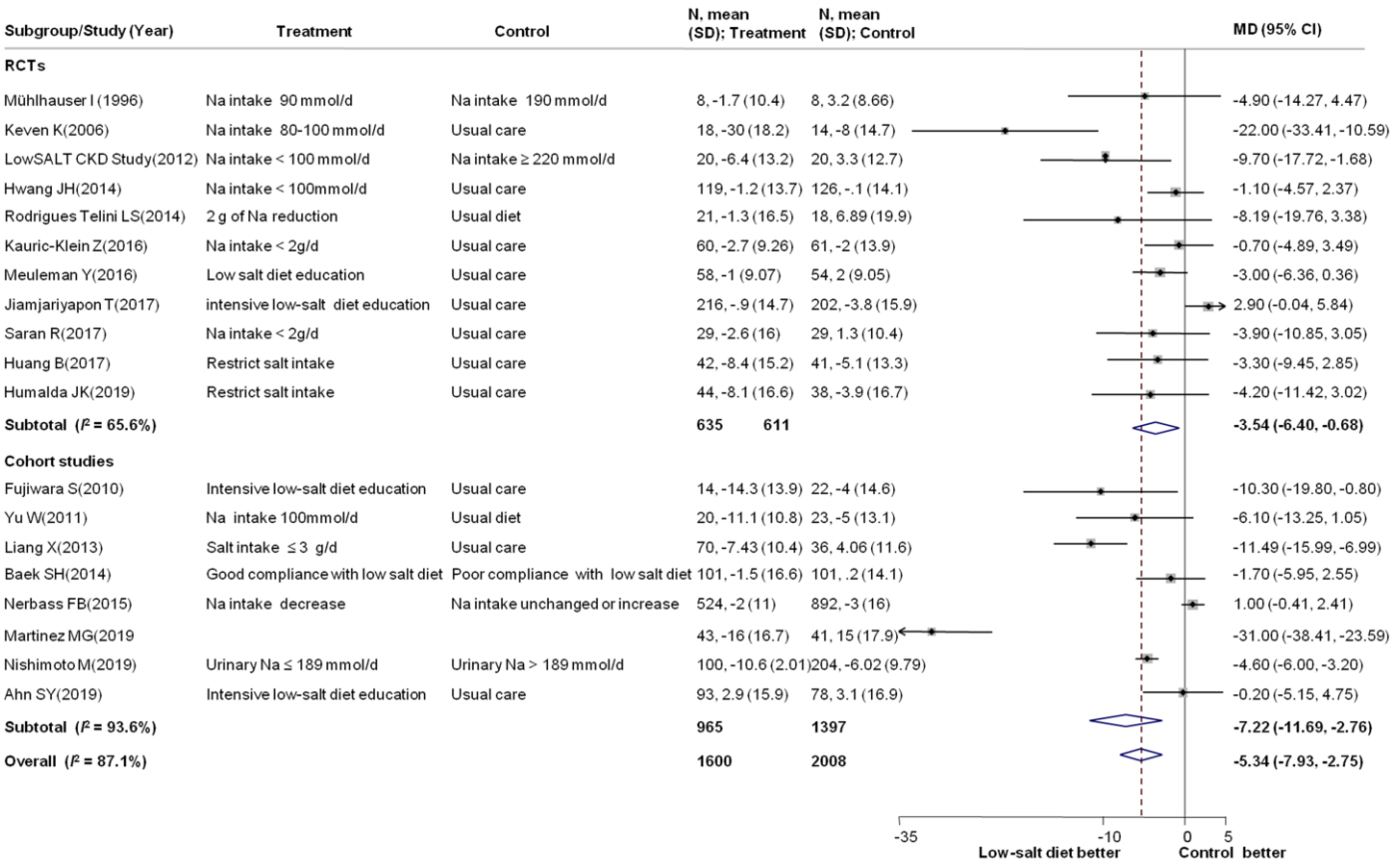
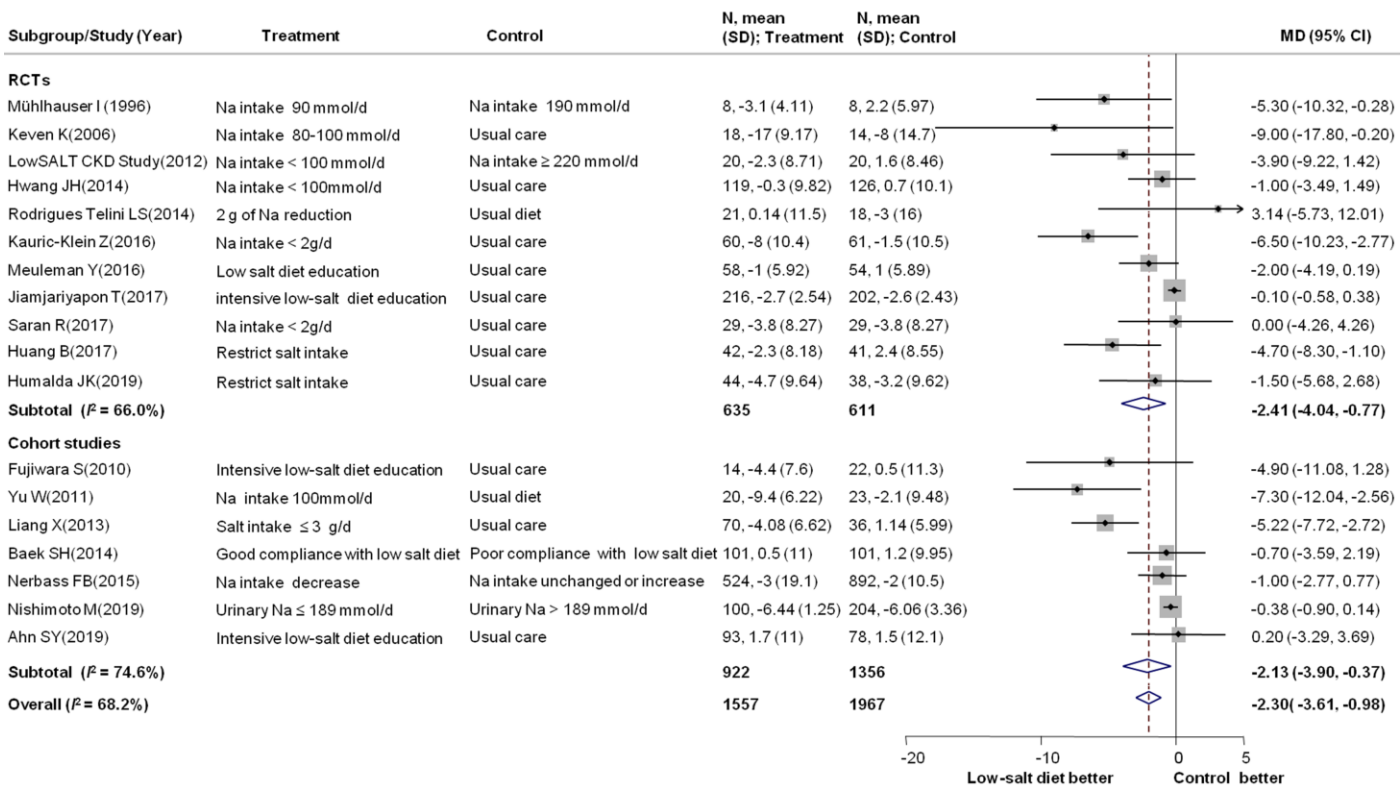


Figure 5 Forest plot for all-cause mortality and cardiovascular events. Cardiovascular events were defined as a composite, including fatal or non-fatal myocardial infarction, fatal or non-fatal stroke, coronary artery revascularization, cardiovascular disease and cardiovascular death. CI, confidence interval; Cr, creatinine; CV, cardiovascular; Na, sodium; N, number of trials; RR, relative risk.



Supplementary file 10 Forest plot for SBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; SBP, systolic blood pressure; SD, standard deviation.



Supplementary file 11 Forest plot for DBP. CI, confidence interval; Cr, creatinine; MD, mean difference; Na, sodium; DBP, diastolic blood pressure; SD, standard deviation.

6. The authors should be discussed about the limitation of combining RCT with cohort studies and discuss more about the subgroup analysis according to study design. I have a concern about pooling evidence from RCT with evidence from cohort studies because cohort studies are prone to selection bias that is usually not found in RCT. Therefore, pooling the results from RCT and cohort studies together will bias the results of meta-analysis.

-Thank you for your advice. We totally agree with you. There is limitation to combine RCTs with cohort studies. We stated this as a part of limitation. See in page 17, line 17-33.

“Cohort studies are prone to selection bias that is usually not found in RCT. Therefore, pooling the results from RCT and cohort studies together will bias the results of meta-analysis. So we performed subgroup analysis according to study design. There was no statistical heterogeneity according to study design for the outcomes. What is noteworthy is that low-salt diet therapy did not produce an beneficial effect on the rate of change in eGFR in RCTs (MD:-0.75; 95% CI: -7.58 to 6.08), but produce beneficial effect on the rate of change in eGFR in cohort studies (MD:2.48; 95% CI: 0.32 to 4.64). ”

Reviewer #3:Thank you for your comments.

VERSION 3 – REVIEW

REVIEWER	Anothaisintawee, Thunyarat Ramathibodi Hospital, Mahidol University, Department of family medicine
REVIEW RETURNED	03-Nov-2021

GENERAL COMMENTS	The term "low salt intervention" should be changed to "low salt diet" because most studies are observational studies that did not provide intervention.

VERSION 3 – AUTHOR RESPONSE

Reviewer #1:

1. The term "low salt intervention" should be changed to "low salt diet" because most studies are observational studies that did not provide intervention.

-Thank you for your advice. The term “low salt intervention” has been changed to “low salt diet”.